Name _____ Period _____ Date _____

2-5 Lesson Quiz: Completing the Square

short form

Question numbers match the Savvas Lesson Quiz. Show your work: the line where you complete the square counts.

2. Write the equation $x^2 + 24x + 60 = 0$ in the form $(x + p)^2 = q$.

Ⓐ $(x + 12)^2 = -60$

Ⓑ $(x + 12)^2 = 84$

Ⓒ $(x + 12)^2 = 204$

Ⓓ $(x + 6)^2 = -24$

Work

3. Solve $0 = x^2 + 6x + 13$ by completing the square.

Ⓐ $x = 5$ and $x = 1$

Ⓑ $x = 3 + i\sqrt{22}$ and $x = 3 - i\sqrt{22}$

Ⓒ $x = -1$ and $x = 1$

Ⓓ $x = -3 + 2i$ and $x = -3 - 2i$

Work

5. Find the zeros of the function $f(x) = -5x^2 + 4x - 1$ by completing the square.

Work

$x = \square + \square\, i$ and

$x = \square - \square\, i$

Scoring: Question 3 and Question 5 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

2-5 Lesson Quiz, short form: answer key

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2. B, $(x + 12)^2 = 84$**.** $x^2 + 24x = -60$; $\left(\frac{24}{2}\right)^2 = 144$; $x^2 + 24x + 144 = -60 + 144$, so $(x + 12)^2 = 84$. Distractors: A adds 144 to the left side only; C adds 144 to +60 without moving the constant; D squares half of 12 instead of half of 24. Savvas quiz item 2, DOK 2, skill A30 (Example 2B; the first three lines of Example 3). **Bonus 5:** 3 for B, 2 for the line that adds 144 to both sides.

3. D, $x = -3+2i$ **and** $x = -3-2i$**.** $x^2+6x = -13$; add $\left(\frac{6}{2}\right)^2 = 9$ to both sides: $(x+3)^2 = -4$; $x+3 = \pm 2i$; $x = -3 \pm 2i$. Savvas quiz item 3, DOK 2, skill A30 (Example 3, Try It 3). **10 points:** 4 for D, 3 for $(x + 3)^2 = -4$ shown, 3 for $x + 3 = \pm\sqrt{-4} = \pm 2i$ and the final pair.

5. $x = \frac{2}{5} + \frac{1}{5}i$ **and** $x = \frac{2}{5} - \frac{1}{5}i$**.** Set $f(x) = 0$ and divide by -5: $x^2 - \frac{4}{5}x + \frac{1}{5} = 0$; $x^2 - \frac{4}{5}x = -\frac{1}{5}$; add $\left(-\frac{2}{5}\right)^2 = \frac{4}{25}$ to both sides: $\left(x - \frac{2}{5}\right)^2 = -\frac{1}{25}$; $x - \frac{2}{5} = \pm\frac{1}{5}i$. Savvas quiz item 5, DOK 2, skill A30 (Example 3 after dividing by a; packet Practice block). **10 points:** 2 for dividing by -5, 2 for adding $\frac{4}{25}$ to both sides, 2 for $\left(x - \frac{2}{5}\right)^2 = -\frac{1}{25}$, 4 for the four boxes (1 each). Accept $0.4 \pm 0.2i$.

Items left off this form (available for a retake or Form B): Savvas item 1, solve $x^2 - 18x + 81 = 4$, select all (answer C and D, $x = 7$ and $x = 11$; Example 1, DOK 2, A30); Savvas item 4, the baking sheet with perimeter 58 in. and area 201.25 in.^{2} (answer A, length 17.5 in. and width 11.5 in.; Example 4, DOK 2, A30: $\ell(29 - \ell) = 201.25$, $(\ell - 14.5)^2 = 9$).

IXL replacement: SmartScore 70 or above on 9MW and NPH replaces this quiz with 20/20.